Week 12: 16–20 November 2026
الأسبوع 12: من 16 - إلى 20 نوفمبر 2026

Subject / الموضوع: Biology **Unit Title / عنوان الوحدة: Module 24: Respiratory & Excretory Systems** **Grade / الصف: 10**

Lessons الحصص	Learning targets أهداف التعلم	Assessment tools and strategies التقييم Formative تكويني/ Summative ختامي / Differentiation التمايز	Resources and links for Parents المصادر و الروابط المخصصة لأولياء الأمور
Session 1 الحصة الأولى M24 L2: Respiratory System الجهاز التنفسي	**DOK 3 – Analyze:** I can analyze how the structure of the respiratory system, especially the alveoli, is adapted for efficient gas exchange, and explain how breathing happens.	2.5 Use multiple media to illustrate concepts (diagrams, videos, models). 2.1 Clarify vocabulary and symbols using visuals, connections, and supports. 5.3 Build fluency with graduated scaffolds and models. **Formative:** labelled respiratory system diagram and Toddle quiz	Kindly check Toddle/Teams for the lesson PPT and practice sheet.
Session 2 الحصة الثانية M24 L3: The Excretory System الجهاز الإخراجي	**DOK 3 – Analyze:** I can analyze how the kidneys filter the blood and form urine, and explain how the excretory system helps maintain homeostasis.	3.2 Highlight big ideas, patterns, and critical features using organizers and examples. 6.3 Facilitate organization with graphic organizers, prompts, and note-taking guides. 7.2 Make learning relevant, authentic, and connected to students' backgrounds. **Formative:** nephron flow chart and exit ticket	Kindly check Toddle/Teams for the lesson PPT and practice sheet.
Homework الواجبات			
Teacher's notes/ Reminders ملاحظات المعلم والتذكير	This week students continue Module 24. Session 1 covers the respiratory system and gas exchange; Session 2 covers the excretory system and how the kidneys form urine. Please encourage your child to review the key vocabulary at home: alveoli, diaphragm, diffusion, gas exchange, kidney, nephron, filtration, reabsorption, excretion.		

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Technology Integration دمج التكنولوجيا	Videos and interactive 3D models of the lungs and kidneys, and Toddle for study-guide practice with second-chance feedback.
Cross-curricular links الروابط بين المواد الدراسية	Physical and Health Education (how exercise changes breathing rate), Chemistry (diffusion and waste products such as urea) and Mathematics (breathing-rate data).
External Assessment Integration دمج التقييمات الخارجية	Aligned to NGSS HS-LS1-2 and HS-LS1-3, and MAP Growth Science (Life Sciences).
Wellbeing/ ATL Skills الصحة النفسية/ أساليب التعلم	Wellbeing: protecting our lungs (no smoking or vaping) and drinking enough water in hot weather. ATL: thinking (linking structure to function) and communication (explaining processes with labelled diagrams and flow charts).
Standards المعايير	• **HS-LS1-2:** Develop and use a model to illustrate the hierarchical organization of interacting systems that provide specific functions within multicellular organisms. • **HS-LS1-3:** Plan and conduct an investigation to provide evidence that feedback mechanisms maintain homeostasis. • **MYP Criterion A:** Knowing and understanding.
Inquiry/ TOK Question البحث والاستقصاء أسئلة TOK	**Inquiry:** How do the respiratory and excretory systems work with other body systems to keep the internal environment of the body stable? **TOK:** Scientists use simple diagrams to model the nephron and the alveoli. How do simplified models help, and limit, our understanding of complex body systems?
Success Criteria معايير النجاح **(Measurable)**	The student has met the learning targets by: لقد حقق الطالب/الطالبة أهداف التعلم من خلال: **Session 1 – M24 L2: Respiratory System** • **MUST (DOK 1):** I can label the parts of the respiratory system (nasal cavity, pharynx, larynx, trachea, bronchi, bronchioles, alveoli, diaphragm) and state the function of each. • **SHOULD (DOK 2):** I can explain how oxygen and carbon dioxide are exchanged in the alveoli by diffusion, and how the diaphragm and rib muscles cause breathing in and out. • **COULD (DOK 3):** I can justify how the features of the alveoli (large surface area, thin walls, many capillaries) increase gas exchange, and predict the effect of smoking or asthma. **Session 2 – M24 L3: The Excretory System** • **MUST (DOK 1):** I can define excretion and identify the organs of the excretory system (kidneys, ureters, bladder, urethra) and other organs that remove wastes (skin, lungs, liver). • **SHOULD (DOK 2):** I can explain the steps of urine formation in the nephron: filtration, reabsorption and secretion.

	• **COULD (DOK 3):** I can justify how the kidneys keep the body's water balance in hot weather, and predict the effects of dehydration or kidney failure.
National Identity Domain **مجالات الربط بالهوية الوطنية**	Belonging الإنتماء Compassion التعاطف Arabic Language اللغة العربية
	المجالArea: Health, safety and wellbeing of the UAE community.
	التطبيقImplementation: • Session 1: students discuss how the UAE protects people's lungs, through laws that ban smoking in public places and through air-quality monitoring during dust storms. • Session 2: students explain why drinking enough water is important in the UAE's hot climate, and discuss how the UAE's national organ donation programme (Hayat) helps people with kidney failure. • Arabic key terms: الجهاز التنفسي (respiratory system), الحويصلات الهوائية (alveoli), تبادل الغازات (gas exchange), الكلية (kidney), النفرون (nephron), الإخراج (excretion).